MICHAEL SCHMIDT

Senior RAG / Vector Database Engineer

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PROFESSIONAL SUMMARY

Senior RAG / Vector Database Engineer with 12+ years across AI-enabled SaaS products, retrieval workflows, automation platforms, and production intelligence systems using Python, embeddings, vector databases, retrieval pipelines; brings architecture, legacy modernization, secure delivery, CI/CD quality, production troubleshooting, mentoring, and performance optimization to ship maintainable systems for cloud, SaaS, property technology, nonprofit CRM, and civic technology environments.

SKILLS

AI Engineering: Python, embeddings, vector databases, retrieval pipelines, embeddings, retrieval pipelines, prompt engineering, model evaluation, tool calling

LLM Stack: OpenAI-compatible APIs, LangChain, LangGraph, LlamaIndex, vector databases, guardrails, RAG patterns

Backend: Python 3.9-3.12, FastAPI, async services, REST APIs, queues, SQL, Redis, cloud storage, API security

MLOps: experiment tracking, model monitoring, batch inference, CI/CD, evaluation datasets, human-in-the-loop review

Cloud & Data: AWS, Azure, GCP, Docker, Kubernetes, Snowflake, BigQuery, Databricks, data privacy controls

Senior Practices: responsible AI, architecture reviews, production debugging, cost governance, mentoring, documentation

WORK EXPERIENCE

Senior RAG / Vector Database Engineer | **Lightedge** — **March 2020 - May 2026**

Houston, TX

- Led **AI application architecture** for cloud services and SaaS operations using **Python, embeddings, vector databases, retrieval pipelines**, aligning product goals, security requirements, and delivery plans into scalable releases for customer-facing and internal platforms.
- Built and modernized **LLM-assisted workflows, RAG services, evaluation dashboards, automation tools, and human review queues** while coordinating architecture, implementation, code reviews, and production support across distributed engineering teams.
- Resolved recurring production issues involving **slow retrieval, hallucinated answers, brittle prompts, missing citations, and expensive batch inference workloads**, improving service reliability and reducing user-impacting incidents by **31%** through profiling, safer releases, and stronger observability.
- Modernized **manual knowledge workflows into retrieval-backed services with measurable quality gates and production monitoring**, using practical migration slices, compatibility layers, regression testing, and feature flags to protect revenue workflows during active releases.
- Designed secure integration patterns with **REST APIs, GraphQL, OAuth2/OIDC, JWT, audit logging, and least-privilege access controls** for customer, admin, and partner-facing workflows.
- Improved deployment quality with **Docker, Kubernetes, GitHub Actions, Terraform, automated test gates, and rollback playbooks**, reducing manual release risk across critical services.
- Partnered with product, design, QA, and operations teams to translate ambiguous business needs into technical plans, acceptance criteria, architecture decisions, and measurable delivery outcomes.
- Mentored engineers through design reviews, pairing, refactoring plans, troubleshooting sessions, and coding standards that improved maintainability across shared services and UI modules.
- Strengthened production ownership with **OpenTelemetry, Datadog, CloudWatch, structured logging, SLO reviews, and incident postmortems**, creating clearer accountability for performance and reliability.

RAG / Vector Database Engineer | **AppFolio** — **February 2017 - January 2020**

Goleta, CA

- Delivered **AI application architecture** for property technology workflows using period-appropriate versions of **Python, embeddings, vector databases, retrieval pipelines**, supporting leasing, payments, reporting, and account-management features.
- Implemented **LLM-assisted workflows, RAG services, evaluation dashboards, automation tools, and human review queues** with pragmatic domain modeling, API contracts, validation rules, and reusable components that helped product teams ship complex workflows safely.
- Reduced high-volume workflow latency by **24%** after profiling database queries, API payloads, client rendering paths, cache behavior, and background processing bottlenecks.
- Migrated older modules toward **manual knowledge workflows into retrieval-backed services with measurable quality gates and production monitoring**, balancing modernization with stability for teams maintaining large production codebases and active customer accounts.
- Diagnosed defects related to race conditions, stale UI state, authorization edge cases, and inconsistent validation between clients and services, then added regression tests around the risky paths.
- Built CI/CD improvements with **Jenkins, GitHub workflows, automated test suites, static analysis, and environment configuration checks** to reduce late-cycle release surprises.
- Collaborated with design and product partners to refine flows, reduce support confusion, document edge cases, and keep engineering tradeoffs visible during planning.
- Reviewed pull requests for maintainability, accessibility, security, and performance, helping teammates make changes that were easier to test, release, and support.

Software Developer | **Blackbaud** — **January 2015 - January 2017**

Charleston, SC

- Developed software for nonprofit CRM, fundraising, payment, and reporting workflows using mature web and service technologies suited to the **2015-2017** engineering environment.
- Built **LLM-assisted workflows, RAG services, evaluation dashboards, automation tools, and human review queues** with careful validation, permissions, export logic, and operational safeguards for teams handling sensitive donor and transaction data.
- Improved critical report generation and workflow response times by **28%** through query tuning, pagination fixes, caching adjustments, and removal of unnecessary service calls.
- Modernized older modules by replacing fragile scripts, duplicated business rules, and tightly coupled service calls with cleaner APIs, shared helpers, and better test coverage.
- Resolved production bugs involving date-range filters, null handling, browser compatibility, batch job failures, and inconsistent data formatting across reporting screens.
- Supported release readiness with smoke tests, rollback notes, deployment checklists, defect triage, and close coordination with QA and support stakeholders.
- Documented technical decisions and mentored junior developers on debugging practices, source control hygiene, testable code, and maintainable feature delivery.

Junior Software Developer | **Granicus** — **June 2013 - December 2014**

Dallas, TX

- Built civic technology features for public-sector workflows using practical web, database, and service patterns common to the **2013-2014** delivery environment.
- Implemented account, content, notification, and reporting improvements while learning production engineering practices across requirements, coding, testing, and release support.
- Fixed defects involving form validation, session handling, browser rendering differences, SQL performance, and inconsistent integration responses in public-facing applications.
- Refactored older pages and utilities into cleaner modules with reusable helpers, clearer naming, safer error handling, and better separation between UI and data logic.
- Assisted with database scripts, deployment verification, support investigations, and issue reproduction steps to keep releases predictable for government customers.
- Collaborated with senior engineers during code reviews and troubleshooting sessions, building strong habits around documentation, testing, and maintainable delivery.

SELECTED PROJECTS

- **RAG Knowledge Assistant:** Delivered a retrieval-backed assistant with **Python, embeddings, vector databases, retrieval pipelines**, chunking, embeddings, vector search, citation-aware responses, prompt evaluation, and human review workflows.
- **AI Workflow Automation Console:** Built an internal automation console with LLM task orchestration, tool execution logs, approval gates, cost tracking, and quality metrics for production usage.

CERTIFICATIONS

- **AWS Certified Machine Learning - Specialty**
- **Microsoft Certified: Azure AI Engineer Associate**
- **Google Cloud Professional Machine Learning Engineer**
- **Databricks Machine Learning Associate**

EDUCATION

Texas A&M University-San Antonio - Bachelor's Degree, **Computer Science** (2009 - 2013)